

CHAPTER 2

Industrial and Systems Engineering

2.1. Introduction

We proposed in Section 1.4 that the principal activity engaged in by engineers that distinguishes them from other professionals is that of the *design of systems*. Basic courses in engineering science that are taken by all engineering students provide a general foundation upon which design principles in particular engineering curricula are later built.

Mechanical engineers design systems that are largely mechanical in nature, such as engines, machines of all types, and transportation vehicles. Electrical engineers design the electrical components of systems, such as integrated circuits and control mechanisms. If M.E.s design mechanical systems, and if E.E.s design electrical systems, can the analogy be extended to claim that I.E.s design industrial systems? *Indeed they do!* Provided, that is, that we define *industrial* to include all types of organizations, not just manufacturing.

2.2. Industrial and Systems Engineering Design

Just what do we mean when we say that industrial and systems engineers *design* industrial systems?¹

Industrial and systems engineers (I.&S.E.s) design systems at two levels. The first level is what Blair and Whitston² have called *human activity systems* and is concerned with the physical workplace at which human activity occurs. The second level is called *management control systems* and is concerned with procedures for planning, measuring, and controlling all activities within the organization.

2.2.1 Human Activity System

The human activity system within an organization consists of the following elements that are designed or prescribed by I.&S.E.s:

- The manufacturing process itself (or the processing procedures of a service organization).
- Materials and all other resources utilized in the production process.
- Machines and equipment.
- Methods by which workers perform tasks.
- Layout of facilities and specification of material flow.
- Material handling equipment and procedures.
- Workplace design.
- Storage space size and location.
- Data recording procedures for management reporting.
- Procedures for maintenance and housekeeping.
- Safety procedures.

2.2.2 Management Control System

The management control system of an organization consists of the following elements that are designed or prescribed by I.&S.E.s:

- Management planning system.
- Forecasting procedures.
- Budgeting and economic analyses.
- Wage and salary plans
- Incentive plans and other employee relations systems.
- Recruiting, training, and placement of employees.
- Materials requirement planning.

¹ At this point, the reader is strongly urged to review the engineering process discussed in Section 1.4 and the definition of industrial engineering presented in Section 1.11.

² R. N. Blair and C. W. Whitston, *Elements of Industrial Systems Engineering* (Englewood Cliffs, N.J.: Prentice Hall, 1971).

- Inventory control procedures.
- Production scheduling.
- Dispatching.
- Progress and status reporting.
- Corrective action procedures.
- Overall information system.
- Quality control system.
- Cost control and reduction.
- Resource allocation.
- Organization design.
- Decision support systems.

Although the elements just described are expressed in manufacturing terminology, the framework is applicable to *any* system.

How well the I.&S.E. performs the above design tasks determines to a very great extent how efficiently and effectively his or her organization will operate. The difficulty involved in performing these design tasks is enormous. Many years of professional practice following a formal educational program in industrial and systems engineering are required before an engineer is capable of understanding how to perform various design tasks and, more important, how these tasks should all be related for an optimal design for the total system.

Chapters 3 through 21 present a quick introduction and overview of some of the major tools that must be mastered by the industrial and systems engineer in order to adequately perform his or her professional activities.

Truthfully speaking, almost all human activity systems are not really *designed*; they simply *evolve*. Few I.&S.E.s really engage in overall system design. Almost all are concerned with *improving* a very small piece of an existing system.

Compared to other engineering disciplines, the design methodology for industrial and systems engineering is relatively primitive and nonrigorous. Much effort needs to be directed toward the development of better design procedures. Even though it can truthfully be argued that the systems with which I.&S.E.s deal are much more complex than purely mechanical or electrical systems, we still have not made satisfactory progress in developing design methodology.

The *teaching* of other engineering disciplines is made relatively easy because the instructor is able to have the students work with an actual example of a physical system. The mechanical engineering instructor can have the students experiment with different fuel mixtures and burn these mixtures in an engine in a laboratory. The electrical engineering instructor can have the students design a circuit in a laboratory based upon fundamental knowledge presented in a lecture.

A professor of industrial and systems engineering cannot bring a production system into a laboratory. It is too big, too complex. Furthermore, it involves human beings who are not only unpredictable and difficult to measure, but who would also be unwilling to be part of a laboratory experiment.

The I.&S.E. student must learn the fundamental nature of the systems he or

she is studying by experimenting with *models* of the systems rather than the actual systems. During the 1940s and 1950s I.E. students typically constructed *scale models* of production systems during their final semester. These scale models, with accompanying engineering drawings, represented the students' "solutions" to particular design problems.

Today, *symbolic models* and *conceptual models* are used rather than physical scale models. Often, these are converted to *computer models* to simulate real-world systems.

Unfortunately, symbolic and conceptual models are very difficult to comprehend. Also, the way these models are typically used in I.&S.E. education does not involve the student in the *design process*. The basic structure of the model is presented by the faculty member and the student merely manipulates independent variables. This exercise is certainly worthwhile, because it provides the student insight into the behavior of a system in response to manipulation of the associated decision variables. The weakness is that the student engages in very little *design activity*.

Obviously, much development work is urgently needed in better methods for teaching design principles to I.&S.E. students. The authors believe that this will receive major attention in the near future.

2.3. Typical I.&S.E. Activities

Different companies expect their I.&S.E.s to perform different kinds of activities. Traditionally, I.E. functions have been concentrated at the *operations* level of a firm. Other firms, recognizing the broad-based skills of their I.E.s, have expanded their activities to include the design of *management systems*. In recent years, as the I.E. role has added a systems flavor, I.&S.E.s are expected to engage in activities at the *corporate* level. To gain an appreciation for the broad spectrum of activities in which an I.&S.E. might be engaged, we present in outline form a list of activities grouped according to the three categories just discussed: production operations, management systems, and corporate services.

2.3.1 Production Operations

A. Related to the *product* or *service*:

1. Analyze a proposed product or service.
 - Determine whether it would be profitable, at various production volumes.
 - Is it compatible with the existing product line?
 - Assess the manufacturability of the design, as prepared by the engineering design department.
 - Determine the best (most cost-effective) material to utilize.
2. Constantly attempt to improve existing products or services.
 - Analyze field data on product usage.
 - Coordinate with the engineering design department on design changes.

3. Perform analyses relating to distribution of the product or delivery of the service.
- B. Related to the *process* of manufacturing the product or producing the service:
 1. Determine the best process and method of manufacture.
 - Apply classification and coding to each component.
 - Apply principles of group technology when appropriate.
 - Apply process planning algorithms to determine production sequence; determine alternate routings.
 - Design and specify appropriate tooling and fixturing.
 2. Select equipment, determine degree of automation, use of robots, and so on.
 3. Balance assembly lines.
 4. Determine the best material flow and material handling procedures and systems.
- C. Related to the *facilities*:
 1. Determine the best layout of equipment.
 2. Determine the appropriate storage facilities for raw materials, work in process, and finished goods inventory.
 3. Determine appropriate preventive maintenance systems and procedures.
 4. Provide for appropriate inspection and test facilities.
 5. Provide adequate utilities for the operation.
 6. Provide for security and emergency services.
- D. Related to *work methods* and *standards*:
 1. Perform work measurement studies; establish time standards; update as required.
 2. Perform methods of improvement studies.
 3. Perform value engineering analyses, eliminating cost and waste to the maximum extent possible.
- E. Related to *production planning and control*:
 1. Forecast the level of activity. (How many units will be sold?)
 2. Analyze the capacity and resource constraints.
 3. Perform operations planning:
 - Facilitate any necessary facility rearrangement.
 - Carry out make-or-buy decisions.
 - Plan production rates; rebalance assembly lines.
 - Construct master production plans.
 4. Perform inventory analysis:
 - Determine current levels of raw materials inventory, work in process, and finished goods inventory.
 - Perform multilevel inventory analyses.
 - Determine appropriate reorder levels, reorder quantities, and safety stock levels.
 - Maintain vendor performance measures on quality, lead time, dependability, and so on.
 - Utilize principles of "just in time," as appropriate.

5. Perform materials requirement planning (MRP).
6. Perform operations scheduling:
 - Allocate resources.
 - Schedule component production.
 - Schedule assembly operations.
 - Design procedures for schedule review and update.
7. Design the quality control system and inspection procedures.
8. Simultaneous smoothing of production, inventory, and work force.
9. Design systems and procedures for shop floor control:
 - Progress and status reporting.
 - Shortage reports.
 - Quality and rework reports.
 - Corrective action procedures.
 - Cost accumulation.
 - Labor utilization reports.
 - Reports on productivity, efficiency, and effectiveness.

2.3.2 Management Systems

A. Related to *information systems*:

1. Determine management information requirements:
 - Identify the decisions that are made by managers at all levels; specify timing of each decision.
 - Determine the specific data/information needed for each decision.
 - Identify the sources of each data element.
 - Determine the preferred form of data and the appropriate media for its transmission.
2. Design the *data base* to support the information system:
 - Specify *input formats* from data sources.
 - Specify the specific *files* required.
 - Within each file, specify the *records* it will contain.
 - For each record, specify the specific data elements (fields) that collectively make up the record.
 - Design the interfaces between the various data files.
3. Design the *management reports* that will be produced:
 - By management level.
 - By time period.
 - Provide for interactive inquiry.
4. Perform *data analyses*, as required.
5. Provide *feedback* to all levels of the organization.
6. Develop and implement *decision support systems*.
7. Analyze the requirements for *data communications* and *computer networks*.

B. Related to *financial and cost systems*:

1. Design a budgeting system.
2. Perform a variety of engineering economy studies.
3. Design, implement, and track cost-reduction programs.
4. Design procedures for systematically updating standard costs.
5. Design systems for generating cost estimates for various purposes.
6. Develop procedures for tracking and reporting cost data for management decision making.

C. Related to *personnel*:

1. Design procedures for employee testing, selection, and placement.
2. Design training and education programs for personnel at all levels in the firm.
3. Design and install job evaluation and wage incentive programs.
 - Individual incentives.
 - Group incentives.
 - Profit-sharing programs.
 - Nonfinancial incentives.
4. Design effective labor relations programs and procedures.
5. Apply the principles of *ergonomics* and *human engineering* to the design of jobs, workplaces, and the total work environment.
6. Develop effective programs of job enhancement.
7. Coordinate the activities of quality circle groups.
8. Design, implement, and monitor effective safety programs.

2.3.3 Corporate Services**A. Relative to *comprehensive planning*:**

1. Design, implement, and monitor a multilevel planning system:
 - Specify mission of organization.
 - Identify key results areas.
 - Specify long-term goals.
 - Determine short-term objectives.
 - Design system for tracking actual results, comparing to plan, and determining corrective action.
2. Assist corporate management in performing strategic planning.
3. Assist corporate management in rationalizing the firm's strategy in the international arena.
4. Perform enterprise modeling:
 - Develop a high-level "business model," in which the major data flows are mapped between the major corporate functions.
 - Employ structured modeling methods to develop a hierarchical breakdown structure of the enterprise functions, subfunctions, and so on.
 - Use the enterprise model to measure business response cycles, determine critical pressure points, and identify strategic opportunities.

5. Perform systems integration activities:
 - Identify key functional interfaces.
 - Determine interdependencies between functions.
 - Rationalize the physical processes; simplify to the extent possible.
 - Rationalize the information interchanges.
 - Implement standard data exchange protocols.
 6. Perform capacity analyses.
 7. Participate in decisions relative to plant expansion and new plant siting.
 8. Provide project management services:
 - Project definition and planning.
 - Work breakdown structures.
 - Network analysis.
 - Project tracking and follow-up.
 9. Assist in implementing the concepts of total quality management throughout the organization.
 10. Provide leadership in resource management:
 - Provide diagnostic services regarding utilization of energy, water, and other resources.
 - Suggest effective means of providing cogeneration capability within the company.
 - Devise effective systems for the management of hazardous wastes, scrap, and other by-products.
 - Provide leadership in constantly seeking means of reducing adverse environmental impacts of the company's products and the processes used in making those products.
- B. Relative to *policies and procedures*:
1. Perform studies relative to organizational analysis and design.
 2. Perform analyses of various functional groupings, recommend improvements to top management.
 3. Develop and maintain policy manuals.
 4. Develop and maintain current procedures relative to all management practices and systems.
- C. Relative to *performance measurement*:
1. Design meaningful performance measures for the key results areas of each organizational unit.
 2. Identify the "critical success factors" or measures of merit for each unit.
 3. Develop methods and systems for analyzing operating data of all units and interpreting the results.
 4. Specify corrective action procedures.
 5. Design reports for all levels of management.
- D. Relative to *analysis*:
1. Analyze systems and construct models:
 - State explicitly the problem being studied.
 - Determine the appropriate solution method.
 - Apply fundamental solution methodologies.
 - Recognize all assumptions pertaining to model and the solution method.

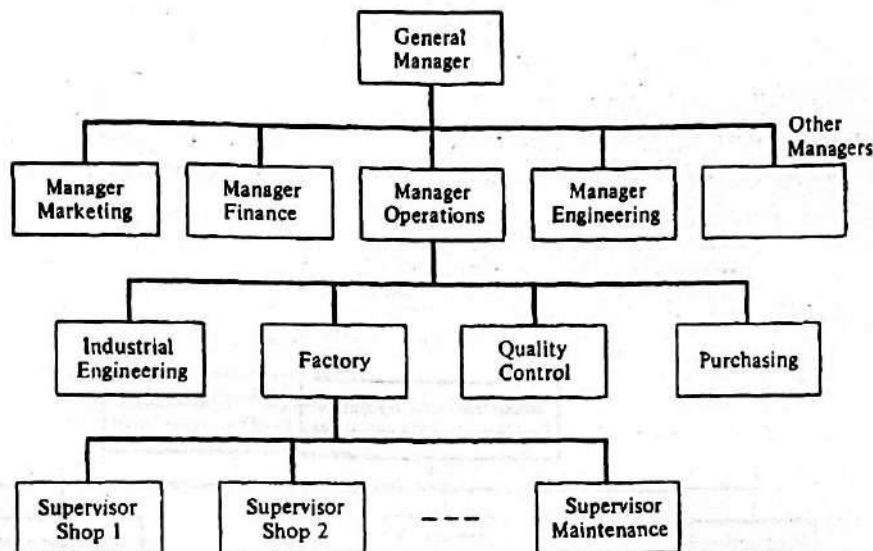


Fig. 2.1. Typical organizational structure for a manufacturing operation.

- Interpret the results of the solution in terms of the original problem statement and the underlying assumptions.
- 2. Perform simulation studies as appropriate.
- 3. Perform operations research studies as appropriate.
- 4. Perform statistical analyses.
- 5. Recognize the dynamic nature of the system being studied and include this feature in proposed solutions.
- 6. Apply the concepts of artificial intelligence and expert systems, as appropriate.
- 7. Conduct designed experiments on appropriate portions of the organization, in an attempt to continuously improve the overall performance of the organization.

One person cannot possibly perform all of the foregoing activities for an organization. I.&S.E. education programs, however, are designed to provide the fundamental principles involved in many of these activities. An individual engineer will specialize in a subset of the foregoing activities. The industrial engineering department will coordinate the individual efforts to address the spectrum of activities outlined.

2.4. Relationship to Total Organization

The various activities listed in the preceding section are not always performed by the industrial engineering department of a company. There are great differences in the scope of activities performed by industrial engineers in different companies. There are also great differences among companies as to where the industrial engineering function fits into the total organizational structure.

Typically, the industrial engineering department is a staff function that reports to the line manager who is in charge of operations. Figure 2.1 illustrates such an organizational arrangement. It is, however, not unusual for the industrial engineering department to be at the "manager" level shown in Figure 2.1.

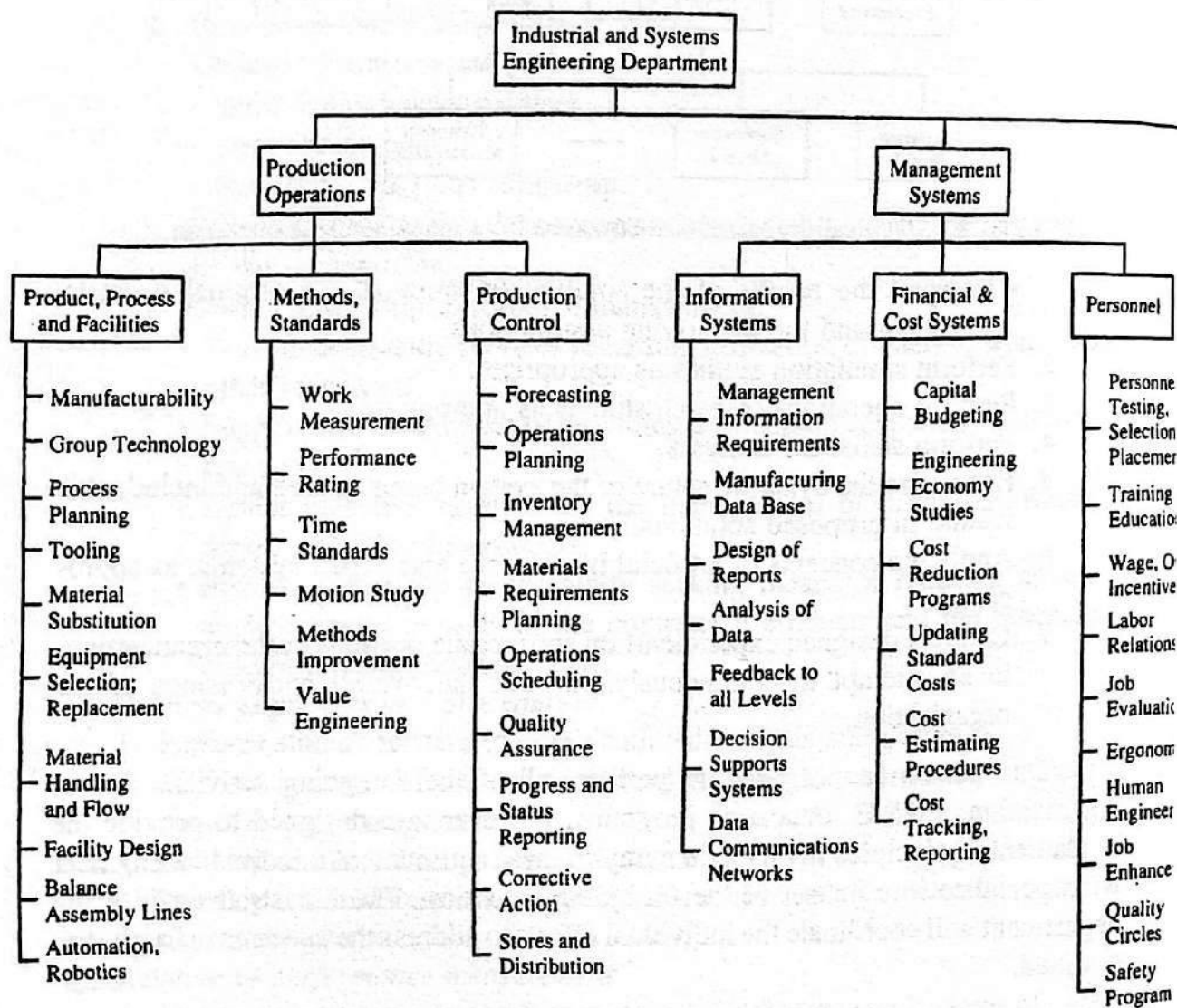


Fig. 2.2. Industrial and systems engineering department organized by major functional grouping

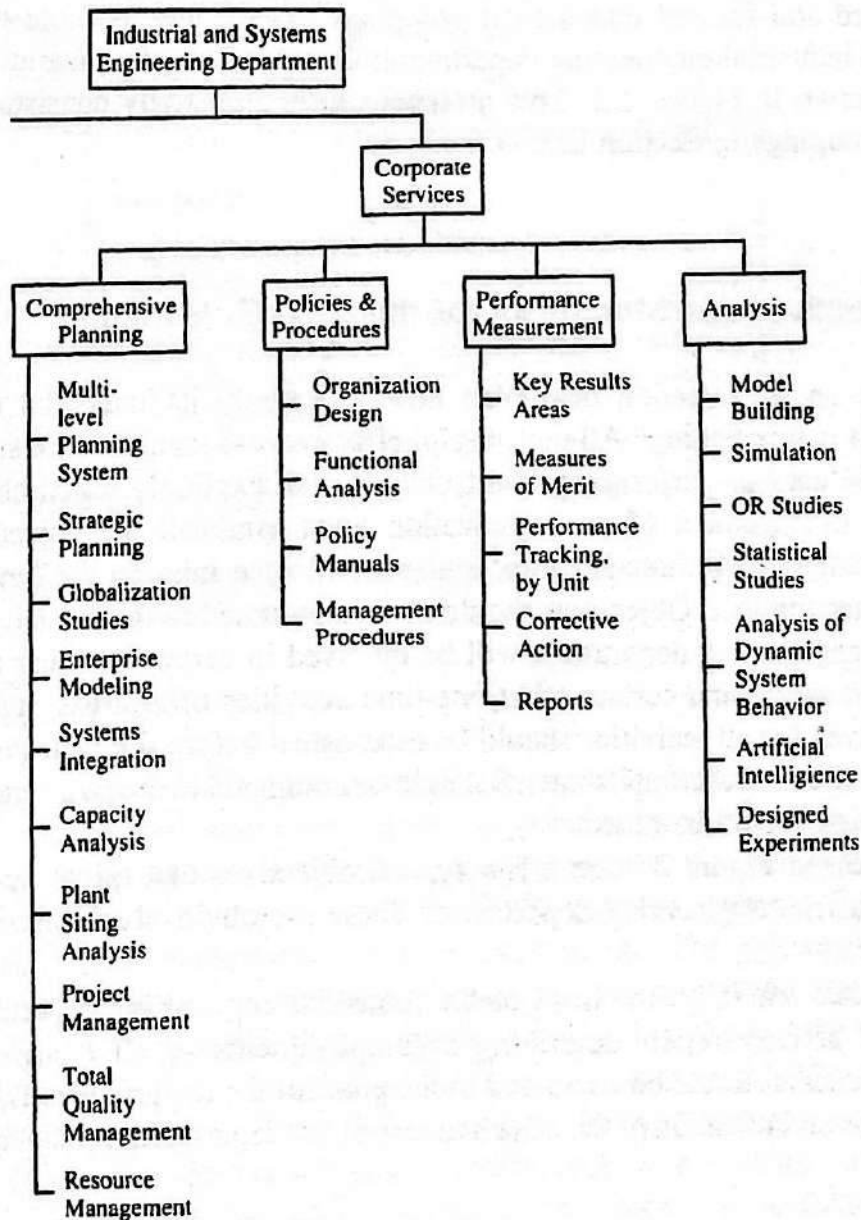


Fig. 2.2. cont.

2.5. Internal Organization of the I.&S.E. Function

For industrial engineering departments that are very small, internal organization is not a problem. Its members simply take on problems as they come and assist each other in their work. Larger departments should be organized into a formal structure.

The internal organizational structure of an industrial engineering department should be *designed* by using basic principles of organizational design. The total scope of activities required to accomplish the industrial engineering mission should be analyzed and formed into logical groupings. These then become the sections within the industrial engineering department. One possible organizational arrangement is shown in Figure 2.2. This arrangement is essentially consistent with the activity groupings in Section 2.3.

2.6. Effectiveness Measures for the I.&S.E. Function

How does an organization determine how effectively its industrial engineering department is functioning? All meaningful effectiveness measures are stated so that they can be used as performance yardsticks against explicitly stated objectives.

The management of an organization must establish the objectives of the industrial engineering function in accordance with the mission the function is expected to accomplish. Objectives should always be stated in measurable terms. The industrial engineering department will be involved in certain ongoing activities of a repetitive nature and certain other one-time activities of a project nature. Clear-cut objectives for all activities should be established before the fact. Periodic measurements of actual accomplishments should be compared to the goals and significant discrepancies should be noted.

Shown in Figure 2.3 are a few typical objectives that might be appropriate for an industrial engineering department. These are obviously for illustrative purposes only.

At least annually, the head of the industrial engineering department should prepare an activity report describing accomplishments of all types. The actual accomplishments should be compared to the goals of the department. This comparison provides an indication of the effectiveness of the department for that time period.

2.7. The Nature of "Systems"³

In previous sections of this chapter, we have used the word "system" many times. We discussed topics such as "human activity systems," "management control

³B. S. Blanchard and W. J. Fabrycky, *Systems Engineering and Analysis*, 2nd ed. (Englewood Cliffs, N.J.: Prentice Hall, 1990).

ON-GOING ACTIVITIES

Achieve cost savings in direct labor equal to three times the departmental budget.

Reduce direct man-hour labor content of production by 10%.

Reduce manufacturing time through improved methods by 7%.

Obtain participation of 50% of workers in cost reduction program.

Reduce the cost of setting time standards by 10%.

Reduce in-process inventory by 8%.

Attain 96% accuracy in management reporting.

PROJECTS

Overhaul unit #3 at a cost not exceeding \$300,000.

Rearrange production equipment to achieve a 15% decrease in flow time; complete project no later than November 15 at a cost not greater than \$18,000.

Fig. 2.3. Typical objectives of an I.&S.E. department.

systems," and "*systems design*." It is useful to define and illustrate several basic concepts associated with "*systems*."

2.7.1 Definitions

A *system* may be defined as a set of components which are related by some form of interaction, and which act together to achieve some objective or purpose. Several words used in this definition need to be explained. *Components* are simply the individual parts, or elements, that collectively make up a system. *Relationships* are the cause-effect dependencies between components. The *objective* or *purpose* of a system is the desired state or outcome which the system is attempting to achieve.

Consider the air-conditioning system in a home. Assume that the system uses a heat pump for both heating and cooling, depending on the need. The *components* of this system include the house (walls, ceiling, floors, furniture, etc.), the heat pump, the thermostat, the air within the system, and the electricity that drives the system. The *relationships* between the system components are as follows:

- (1) The *air temperature* depends on:
 - (a) Heat transfer through the walls, ceiling, floor, and windows of the house.
 - (b) Heat input or output due to heat pump action.
- (2) The *thermostat* action depends on:
 - (a) Air temperature.
 - (b) Thermostat setting.

(3) The *heat pump* status depends on:

- (a) Thermostat action.
- (b) Availability of electricity.

Although there are other relationships in this system, the ones listed above serve to give physical meaning to the terms we are using. We will look in greater detail at this system in a later section.

2.7.2 System Classifications

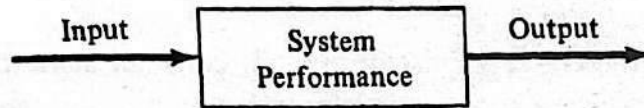
Systems may be classified in a number of different ways. On the following pages we discuss a few classifications that illustrate the similarities and dissimilarities of systems.

- *Natural vs. Man-Made Systems*—*Natural systems* are those that exist as a result of processes occurring in the natural world. A river is an example of a natural system. *Man-made systems* are those that owe their origin to human activity. A bridge built to cross a river is an example of a man-made system.
- *Static vs. Dynamic Systems*—A *static system* is one that has structure but no associated activity. The bridge crossing a river is a static system. A *dynamic system* is one that involves time-varying behavior. The U.S. economy is an example of a dynamic system.
- *Physical vs. Abstract Systems*—A *physical system* is one that involves physically existing components. A factory is an example of a physical system, since it involves machines, buildings, people, and so on. *Abstract systems* are those in which symbols represent the system components. An architect's drawing of a factory is an abstract system, consisting of lines, shading, and dimensioning.
- *Open vs. Closed Systems*—An *open system* is one that interacts with its environment, allowing materials (matter), information, and energy to cross its boundaries. A *closed system* operates with very little interchange with its environment.

2.8. Feedback Control in Systems

Essentially all systems exist to perform some function. A watch is designed to keep track of time. We set the watch and then the internal mechanism causes the hands to move. However, as time goes on, the time portrayed on the face of the watch may be different from the true time, due to imperfections in the mechanism. The watch is called an *open-loop system* because it is not able to provide for its own control or modification. Figure 2.4 shows a diagram of an open-loop system and some of its characteristics.

Many systems that we encounter are able to control or adjust their own



- Not aware of its own performance
- Past action has no influence on future action
- Possesses no means to provide for its own control or modification
- $\text{Output} = f(\text{input})$, but $\text{input} \neq f(\text{output})$

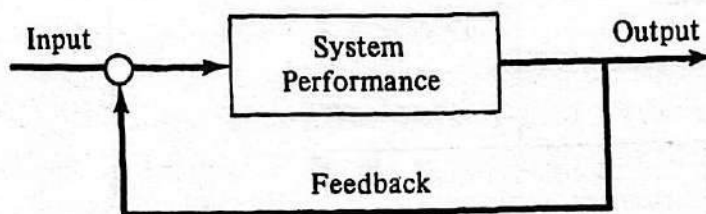
Examples:

- Toaster – doesn't know whether it burned the toast
- Watch – not self-correcting
- Auto without a driver

Fig. 2.4. Open-loop system.

performance. Such systems are called *closed-loop systems*. Specifically, a closed-loop system is one that controls or adjusts its own performance in response to output data generated by the system itself. The home air-conditioning system described in Section 2.6 is a closed-loop system. Figure 2.5 shows a diagram of a closed-loop system and some of its characteristics.

The only difference between the diagrams of Figures 2.4 and 2.5 is the presence of a “feedback” loop in Figure 2.5. *Feedback* can be defined as the system function that obtains data on system performance (output), compares the actual performance to the desired performance (a *standard* or a *criterion*) and determines the modifications (corrective action) necessary prior to the next execution of system performance. This definition of feedback is long and complex. Indeed, it is common to treat this subject in a semester-long university course, usually in electrical or mechanical engineering curricula.



- Aware of and influenced by its own past performance
- Results of past action influences future action
- Senses its performances and automatically makes adjustments
- $\text{Output} = f(\text{input})$ and $\text{input} = g(\text{output})$

Examples:

- Toaster and operator – adjust to desired darkness
- Watch and owner – adjust time to “standard”
- Auto with a driver

Fig. 2.5. Closed-loop system.

To illustrate the concepts just discussed, consider again the home air-conditioning system, in which a heat pump is used for both heating and cooling. For this illustration, we will consider only the heating cycle. It is desired to maintain a temperature of 70°F. To avoid having the unit cycle on and off too frequently, the thermostat is set to allow fluctuations between 67 and 73°F. Such a closed-loop feedback system can be illustrated as shown in Figure 2.6. The system illustrated is somewhat simplified. For example, the decision process would probably protect against turning the unit on if there was no power. It might also have some time delays built in relative to the frequency of switch activation.

Our interest in the concept of feedback is a very specialized one. At the beginning of this chapter, we established that industrial and systems engineers design systems at two levels. The first level, *human activity systems*, is concerned with the physical environment in which human activity occurs. The second level, *management control systems*, is concerned with procedures for planning, measuring, and controlling all activities within an organization.

When we design management control systems, we must explicitly provide for information feedback to the decision processes that control and adjust ongoing activities. For example, in a production process, we establish a production plan that specifies the desired outputs to be achieved through time. We then implement the plan. After a short period of operation (say, one week), we measure actual outputs and compare them to the desired outputs (the *plan* or *standard*). If the actual outputs are within an acceptable range from the desired outputs, we make no modifications

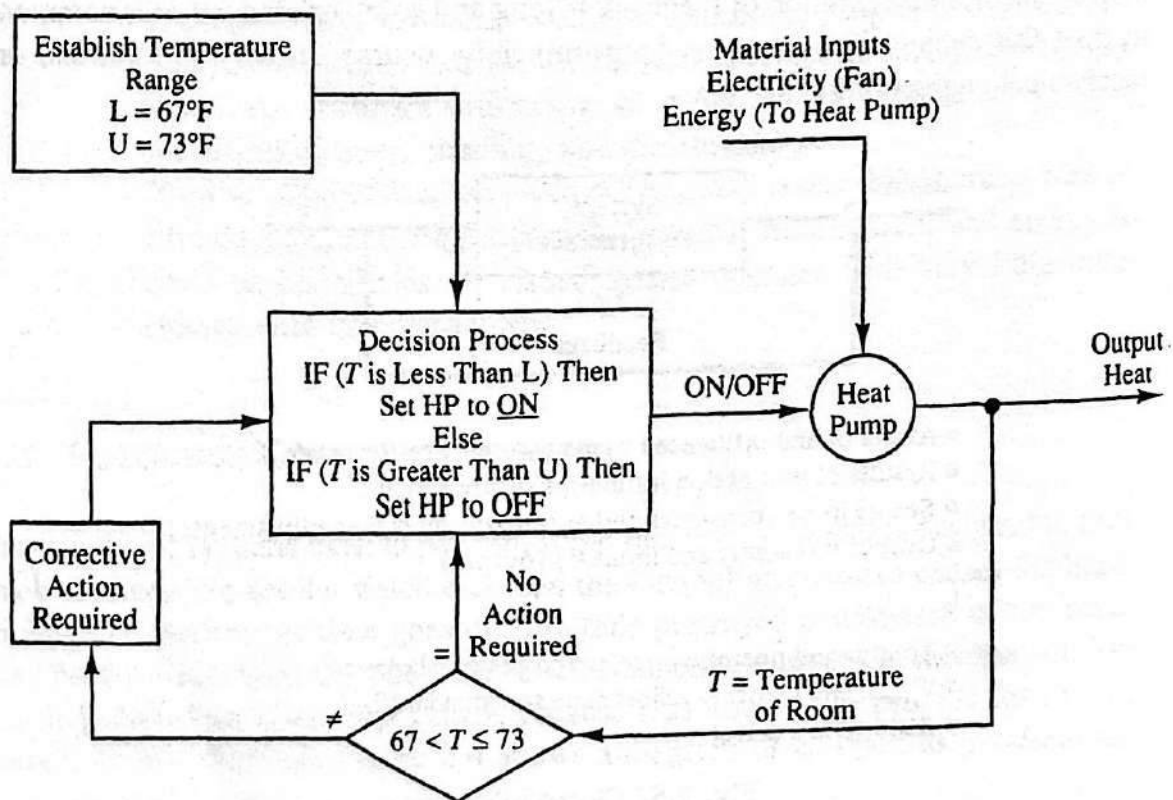


Fig. 2.6. Feedback-control system for heat pump system; heating cycle only.